

PRINT NAME: _____ ID#: _____

- Show your work. Answers with no or incorrect steps will not receive partial credit.
- The exam is self-explanatory. The instructor will not interpret the questions.

1. (4 points) Let the linear spaces $X = C^1([0, 1])$ and $Y = C([0, 1])$ be both endowed with norm $\|x\| := \max_{0 \leq t \leq 1} |x(t)|$. Define $T : X \rightarrow Y$ by $(Tx)(t) = x'(t)$ (x' the derivative of x) for all $t \in [0, 1]$ and $x \in X$. Prove that

- T is a closed operator.
- T is not a continuous operator.

Proof. (a) Let $\{x_k\}$ be a sequence in X such that $x_k \rightarrow x$ in X and $Tx_k \rightarrow y$ in Y . Since the convergence in the sense of $\|\cdot\|$ is uniform convergence, we know

$$x_k(t) - x_k(0) = \int_0^t x'_k(s) ds \rightarrow \int_0^t y(s) ds, \quad \forall t \in [0, 1],$$

due to $Tx_k = x'_k \rightarrow y$ in Y . On the other hand, we know

$$x_k(t) - x_k(0) \rightarrow x(t) - x(0), \quad \forall t \in [0, 1],$$

due to $x_k \rightarrow x$ in X . Hence $x(t) - x(0) = \int_0^t y(s) ds$ for all $t \in [0, 1]$, which implies that $x \in X$ and $Tx = y$. Hence T is closed.

- Let $x_k(t) := \sin(kt)/k$ for all $k \in \mathbb{N}$. Then we can show that $\|x_k\| = 1/k \rightarrow 0$ as $k \rightarrow \infty$ and $\|Tx_k\| = \|x'_k\| = 1$ for all $k \in \mathbb{N}$. Hence T is not continuous.

□

2. (4 points) Let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space and M a linear subspace of X . Recall that $M^\perp := \{y \in X : \langle x, y \rangle = 0, \forall x \in M\}$. Prove that

- $M^\perp$ is a closed linear subspace of X .
- $\bar{M} = M^{\perp\perp}$ where $M^{\perp\perp} := (M^\perp)^\perp$. [Hint: notice that $\bar{M}^\perp \subset M^{\perp\perp}$.]

Proof. (a) It is easy to show that $M^\perp$ is a linear subspace of X . Let $\{y_k\}$ be a sequence in $M^\perp$ such that $y_k \rightarrow y$ for some $y \in X$. Then for any $x \in M$, we have $\langle x, y_k \rangle \rightarrow \langle x, y \rangle$ which implies $\langle x, y \rangle = 0$ since $\langle x, y_k \rangle = 0$ for all $k \in \mathbb{N}$. Hence $y \in M^\perp$ and thus $M^\perp$ is closed.

- For any $x \in M$ and any $y \in M^\perp$, we have $\langle x, y \rangle = 0$. Hence $x \in M^{\perp\perp}$. Therefore $M \subset M^{\perp\perp}$. Since $M^{\perp\perp} := (M^\perp)^\perp$ is closed due to Part (a), we know $\bar{M} \subset M^{\perp\perp}$.

If there exists $x \in M^{\perp\perp} \setminus \bar{M}$, then there exist unique $y \in \bar{M}$ and nonzero $z \in \bar{M}^\perp$ such that $x = y + z$. Hence $\langle x, z \rangle = \|z\|^2 > 0$. However $z \in \bar{M}^\perp \subset M^\perp$ and $x \in M^{\perp\perp}$ which yield $\langle x, z \rangle = 0$, a contradiction. Hence $M^{\perp\perp} \subset \bar{M}$.

Combining the two results above yields $\bar{M} = M^{\perp\perp}$.

□

3. (4 points) Let X be a Banach space, E a nonempty open subset of X , and F a subset of first category in X . Show that $E \setminus F$ is dense in E . [Hint: It suffices to show $E \subset \overline{E \setminus F}$. To this end, consider to prove by contradiction while noticing that nonempty open sets are of second category.]

Proof. We only need to show $E \subset \overline{E \setminus F}$. If not, then $E \setminus \overline{(E \setminus F)}$ is nonempty and open, and thus of second category. On the other hand,

$$E \setminus \overline{(E \setminus F)} \subset E \setminus (E \setminus F) \subset F,$$

which contradicts to F being of first category.

□

4. (4 points) Let X and Y be Banach spaces, $T_k \in L(X, Y)$ for all $k \in \mathbb{N}$. For any $x \in X$, the sequence $\{T_k x\}$ is convergent in Y . Show that there exists $T \in L(X, Y)$ such that $T_k x \rightarrow Tx$ for any $x \in X$ and $\|T\| \leq \liminf_{k \rightarrow \infty} \|T_k\| < \infty$. [Hint: use uniform boundedness theorem to show $\liminf_{k \rightarrow \infty} \|T_k\| < \infty$.]

Proof. We just define $T : X \rightarrow Y$ by $Tx := \lim_{k \rightarrow \infty} T_k x$ for any $x \in X$. It is easy to show that T is linear. Furthermore,

$$\|Tx\| = \lim_{k \rightarrow \infty} \|T_k x\| = \liminf_{k \rightarrow \infty} \|T_k x\| \leq \liminf_{k \rightarrow \infty} \|T_k\| \|x\|,$$

which implies that $\|T\| \leq \liminf_{k \rightarrow \infty} \|T_k\|$. By uniform boundedness theorem, we know $\liminf_{k \rightarrow \infty} \|T_k\| < \infty$. Hence $T \in L(X, Y)$. $\square$

5. (4 points) Let X be a Hilbert space and $T : X \rightarrow X$ be linear and self adjoint, i.e., $\langle Tx, y \rangle = \langle x, Ty \rangle$ for all $x, y \in X$. Show that T is continuous. [Hint: use closed graph theorem.]

Proof. Let $\{x_k\}$ be a sequence in X such that $x_k \rightarrow x$ and $Tx_k \rightarrow y$. Then for any $z \in X$, we have $\langle Tx_k, z \rangle \rightarrow \langle y, z \rangle$ and

$$\langle Tx_k, z \rangle = \langle x_k, Tz \rangle \rightarrow \langle x, Tz \rangle = \langle Tx, z \rangle.$$

Hence $\langle y, z \rangle = \langle Tx, z \rangle$. As z is arbitrary, we know $y = Tx$, which means that T is closed. By closed graph theorem, we know T is continuous. $\square$

6. (4 points bonus) Mark each of the following assertions True (T) or False (F).

- (a) ☐ The set $\{x \in \mathbb{R} : \sin(x) = 0\}$ is of first category in $\mathbb{R}$.
- (b) ☐ The set $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is of second category in $\mathbb{R}^2$.
- (c) ☐ Let X and Y be Banach spaces and $T : X \rightarrow Y$ be linear. Then T is bounded if and only if T is continuous at $0 \in X$.
- (d) ☐ Let X and Y be Banach spaces and $T \in L(X, Y)$. If T is surjective, then there exists $\delta > 0$ such that $U(0; \delta) \subset T(B(0; 1))$ where $B(0; r)$ and $U(0; r)$ denote the open balls centered at origin with radius r in X and Y respectively.

Solution.

- (a) True. The interior of the closure is empty.
- (b) False. The interior of the closure is empty.
- (c) True. T is bounded iff T is continuous iff T is continuous at 0 .
- (d) True. Open mapping theorem.